Alberto Progida

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Personal Profile

Spacecraft engineer with hands-on experience across the full hardware stack (mechanisms, composite structures, avionics, harnessing, and AIT), comfortable owning problems end-to-end in fast-moving teams. Previously, a first engineer at a two-founder ISAM robotics startup, where I built a 3-DOF demonstrator in under two months that helped raise a £70k pre-seed round and secure the team's first letters of intent. Most recently at Oxford Space Systems, I owned mechanism design, thermoelastic analyses and PDR design reviews for a deployment and gimballing system and participated in AIT and environmental campaigns. I thrive working across disciplines and in small, committed teams: taking ownership of consequential decisions and executing with technical rigour and clear communication.

Engineering Experience

OSS - Oxford Space Systems Ltd. *Oxford, England, UK*

Associate Spacecraft Mechanical Engineer *Sept 2024 - Current*

- Owned the design and kinematic modelling of a **gimballed antenna** motorisation chain to **ESA ECSS standards** using DC and Frameless BLDC motors, bevel gears, leadscrews and harmonic drives from concept to DM test,
- Planned and executed **DM-level testing**, implementing design modifications to reduce assembly time and engaging with suppliers to reduce procurement costs,
- Secured buy-in from Chief Engineering and **funding agencies** by leading trade-offs and presenting technical design packages at **PDR-level reviews**,
- Pre- and post-processed system-level thermoelastic, static and modal NASTRAN analyses via **Simcentre FEMAP**,
- 6-month clean-room based AIT activities for a high-stakes, multi-million pound **test campaign of EQM hardware** from piece parts to qualified system,
- Specification and execution of TVAC/Environmental/Vibration tests both internally and externally.

Lodestar Ltd. *London, England*

Robotics Engineer *July 2023 - Sept 2023*

- Designed, built, tested a **scaled 3-DOF demonstrator** in <2 months helping to raise an initial £70k pre-seed round from Entrepreneurs First (EF),
- Designed, sized and assembled a modular powertrain of BLDC motors and Harmonic Drives,
- Developed a parametric preliminary design baseline for a 7-DOF LEO-capable robotic arm detailed with power, thermal, packaging, and cost budgets used for early customer outreach and to **secure the team's first LOIs**.

ORBEX - Orbital Launch Express Ltd. *Forres, Scotland*

Structures Engineer *Sept 2022 - July 2023*

- Owned the DFM of metallic/composite flight hardware, including **tolerance stack-up and GD&T to BS8888**,
- Designed and analysed metallic pressure vessel components and **bolted & fluid connections operating at cryo**,
- Liaised with machinists and assembly technicians, leading meetings to **collaborate on clear and safe AIT procedures**,
- Managed and curated **>3000-part production-quality CAD assemblies**.

Flowcopter Ltd. *Edinburgh, Scotland*

Avionics Hardware R&D Intern *June 2022 - Sept 2022*

- **Hardware reverse engineering** of ROTAX light aircraft ECU (Engine Control Unit) for autonomous operation,
- Design, component selection and test of **multi-layer PCBs** for data conditioning and high-power switching up 500W,
- Hardware-level integration of Data Acquisition, Engine and Hydraulic Pump Controllers and Autopilot,
- Instrumentation and control **wire harness planning** & procurement.

RFA - Rocket Factory Augsburg AG *Augsburg, Germany*

Hardware Test Engineering Intern *June 2021 - Aug 2021*

- Wiring, testing and **commissioning of instrumentation** on a major MW-scale turbo-machinery test rig,
- Conceptualisation, design and operation of pneumatic test rig for acceptance testing of COTS valves,
- Read, created and interpreted P&IDs for hydrostatic testing and safe operation of high pressure systems.

Education

Imperial College London *London, UK*

MSc Advanced Aerospace Engineering - **Distinction - 85.6% - Second of my cohort** *Sept 2023 - Sept 2024*

- **Thesis**: FEA, Manufacture and Characterisation of a Composite Flexure with Integrated Electrical Connections
- **Relevant Courses**: Spacecraft Structures, Systems Engineering, Composite Structures, Advanced Propulsion

University of Edinburgh *Edinburgh, Scotland*

BEng Mechanical Engineering (Hons) - **First Class - 77%** *Sept 2019 - Jul 2022*

- **Dissertation**: Design of a Subsea Actuator for Sensing Applications in Energetic Tidal Channels — 86%
- Institution of Mechanical Engineers (IMechE) Best Student Award

United World College of the Atlantic *St. Donat's Castle, Wales*

International Baccalaureate - **Full ride scholarship, 41/45, top 2%** *May 2017 - May 2019*

Projects

Composite Tape Spring with Integrated Electrical Connections *London, England*

Structural Meta-Materials Research Group - Imperial College London *Jan 2024 - Sept 2024*

- MSc Thesis supervised by Dr. Matthew Santer, now published in the *CEAS Space Journal*
- Experimenting with **ultra-thin printed circuits** (<25um) for lamination within a high-strain CFRP flexure
- Development of novel **layup techniques** to investigate the impact of the circuits on flexure performance
- **Optical and Scanning Electron microscopy** to investigate layup quality and identify defects
- Built and successfully correlated an **Abaqus Explicit Dynamic model** with experimental deployments

Chief Engineer - Darwin III Sounding Rocket Program *Edinburgh, Scotland*

endeavour Student Rocketry Team *Sept 2021 - Oct 2022*

- **Led 40+ students** in the design, manufacturing and successful **launch of a supersonic sounding rocket,**
- Collaborated with subsystem leads to specify technical requirements achievable within our timelines and **£25k budget.**
- Successfully launched the 34kg vehicle to 7000m, reaching a **top speed of Mach 1.7**
- Used **Altium Designer to create and test 10+ PCBs** throughout the Avionics Bay and Payload Section with analog and digital design, RF sections and serial comms,
- Hands-on manufacture of carbon- and glass-fiber composite structures **(Filament Winding, Pre-Preg and Wet Layup)**.

Skills & Interests

Mechanical	GD&T (BS8888) · 3D Printing · FEMAP Thermal · Abaqus (Explicit/Standard) · FEMAP · SolidWorks · Composite layup & processing · DFM · AIT operations · Manual Machining · Hand Calculations
Electrical	KiCad · Altium Designer · PCB design · Schematics · Harness design · High-power switching · Board Bring-up
Programming	Claude Code · Python (Pandas, NumPy, SciPy, Numerical Methods, GUIs, Data Visualisation) · Matlab · PowerBI, LaTeX
Interests	Long distance running, sailing, hiking, hard sci-fi literature & fresh pasta

Achievements

2026	**Published Article**, Integrated electrical connections in deployable flexures *(link to paper)*	*London, UK*
2022	**Best Student**, Institution of Mechanical Engineers (IMechE)	*Edinburgh, UK*
2022	**First Prize**, European Rocketry Challenge: Best Payload - Awarded by ESA	*Ponte de Sor, PT*
2022	**BrightSpark**, Promising Young Engineer Award *(link to RS Comp./Electronics Weekly)*	*IET London, UK*
2021	**First Prize**, Mach 21: Best CanSAT	*Machrihanish AFB, UK*
2017	**4th Place Worldwide**, Global ZeroRobotics Competition	*ESTEC Noordwijk, NL*

Languages

Italian	Native proficiency
English	Bilingual proficiency
French	Conversational B1